화학수학

다섯 번째 수업

다변수 적분

급수

Vector Derivatives

- Gradient (기울기 연산자) ∇f

  scalar → vector

- Divergence (발산 연산자) $\nabla \cdot \vec{F}$

  vector → scalar

- Curl (회전 연산자) $\nabla \times \vec{F}$

  vector → vector

- Laplacian (라플라스 연산자) $\nabla^2 f$

  scalar → scalar (2nd order)

Vector Derivatives

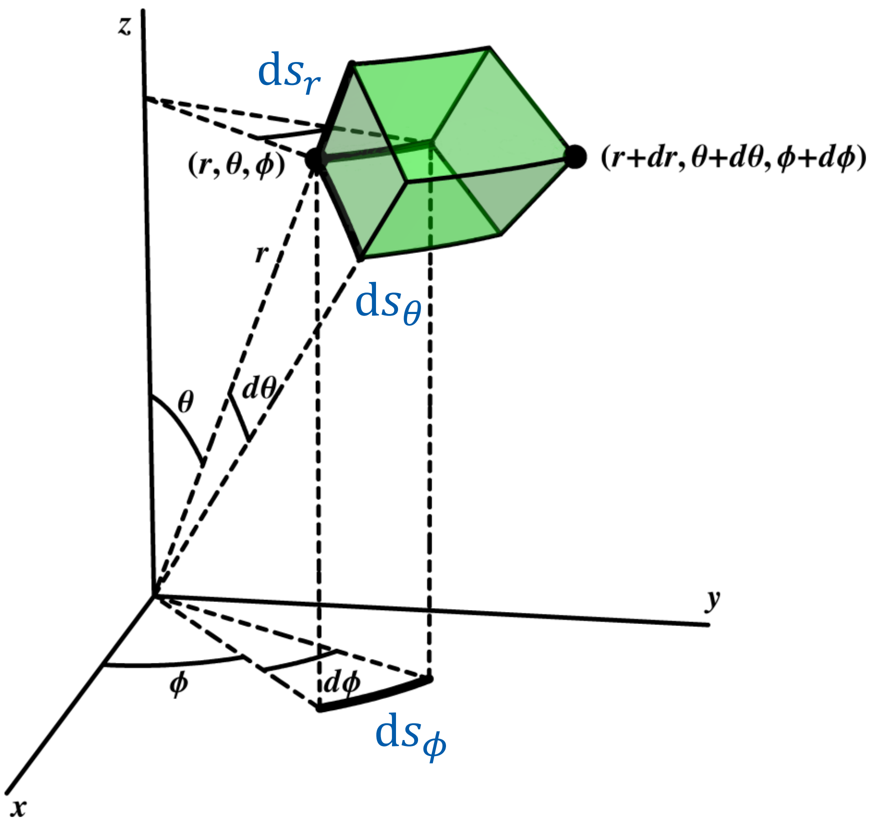
$$\nabla f = \hat{\imath}\left(\frac{\partial f}{\partial x}\right) + \hat{\jmath}\left(\frac{\partial f}{\partial y}\right) + \hat{k}\left(\frac{\partial f}{\partial z}\right)$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x}\right) + \left(\frac{\partial F_y}{\partial y}\right) + \left(\frac{\partial F_z}{\partial z}\right)$$

$$\nabla \times \vec{F} = \hat{\imath}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2}\right) + \left(\frac{\partial^2 f}{\partial y^2}\right) + \left(\frac{\partial^2 f}{\partial z^2}\right)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r\, d\theta$$

$$ds_\phi = r \sin\theta\, d\phi$$

(Image: https://bridge.math.oregonstate.edu/Book/dscoord.html)

Infinitesimal Displacements

In Cartesian coordinates,

$$\mathrm{d}\vec{r} = \hat{\imath}\,\mathrm{d}x + \hat{\jmath}\,\mathrm{d}y + \hat{k}\,\mathrm{d}z$$

In spherical polar coordinates,

$$\mathrm{d}\vec{r} = \hat{e}_r \mathrm{d}r + \hat{e}_\theta r\,\mathrm{d}\theta + \hat{e}_\phi r \sin\theta\,\mathrm{d}\phi$$

In general,

$$\mathrm{d}\vec{r} = \hat{e}_1 h_1 \mathrm{d}q_1 + \hat{e}_2 h_2 \mathrm{d}q_2 + \hat{e}_3 h_3 \mathrm{d}q_3$$

Vector Derivatives in Other Coordinates

$$\nabla f = \hat{e}_1 \frac{1}{h_1}\frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2}\frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3}\frac{\partial f}{\partial q_3}$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(F_1 h_2 h_3)}{\partial q_1} + \frac{\partial(F_2 h_1 h_3)}{\partial q_2} + \frac{\partial(F_3 h_1 h_2)}{\partial q_3}\right]$$

$$\nabla \times \vec{F} = \hat{e}_1 \frac{1}{h_2 h_3}\left[\frac{\partial(h_3 F_3)}{\partial q_2} - \frac{\partial(h_2 F_2)}{\partial q_3}\right] + \hat{e}_2 \frac{1}{h_1 h_3}\left[\frac{\partial(h_1 F_1)}{\partial q_3} - \frac{\partial(h_3 F_3)}{\partial q_1}\right]$$

$$+ \hat{e}_3 \frac{1}{h_1 h_2}\left[\frac{\partial(h_2 F_2)}{\partial q_1} - \frac{\partial(h_1 F_1)}{\partial q_2}\right]$$

Vector Derivatives in Other Coordinates

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right]$$

Line Integral (선적분)

Consider a differential with two independent variables:

$$\mathrm{d}F = M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y$$

For the curve C joining (x_0, y_0) and (x_1, y_1),

the line integral (path integral) is defined as

$$\int_C \mathrm{d}F = \int_C [M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y]$$

Evaluation of Line Integral

$$\int_C \mathrm{d}F = \int_C [M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C \mathrm{d}F = \int_C [M\{x, y(x)\}\mathrm{d}x + N\{x(y), y\}\mathrm{d}y]$$

$$\int_C \mathrm{d}F = \int_{x_0}^{x_1} M\{x, y(x)\}\,\mathrm{d}x + \int_{y_0}^{y_1} N\{x(y), y\}\,\mathrm{d}y$$

Line Integrals of Exact Differentials

If $\mathrm{d}F$ is an exact differential, the line integral $\int_C \mathrm{d}F$

- depends only on the initial and final points.
- equals to F(final point) $-$ F(initial point).

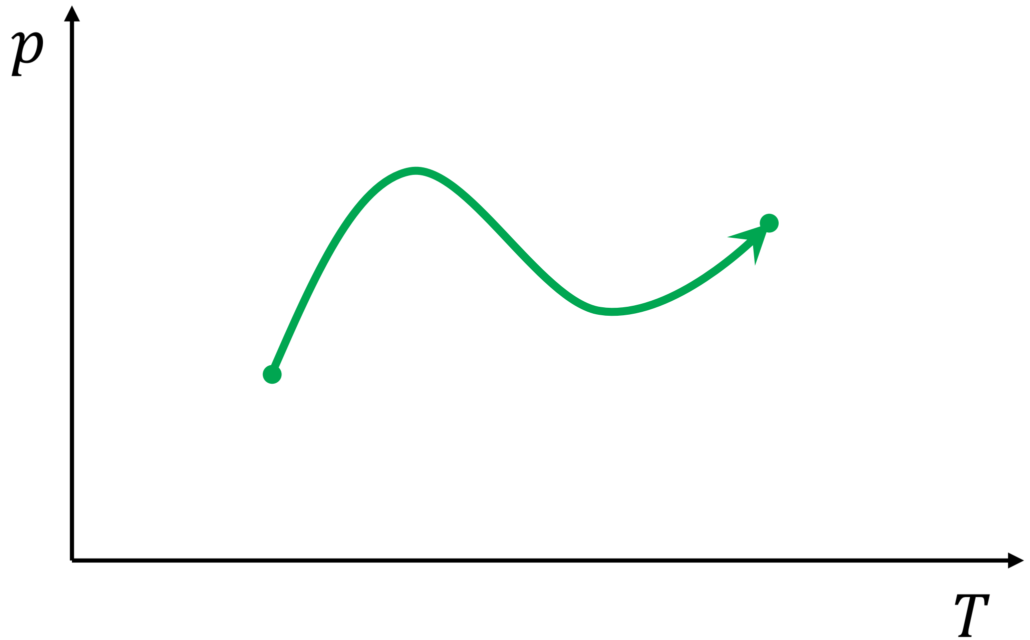
For an exact differential $\mathrm{d}F$ of the function $F(x, y)$,

with the initial point (x_0, y_0) and the final point (x_1, y_1),

$$\int_C \mathrm{d}F = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y \mathrm{d}x + \left(\frac{\partial F}{\partial y} \right)_x \mathrm{d}y \right] = F(x_1, y_1) - F(x_0, y_0)$$

Line Integrals in Thermodynamics

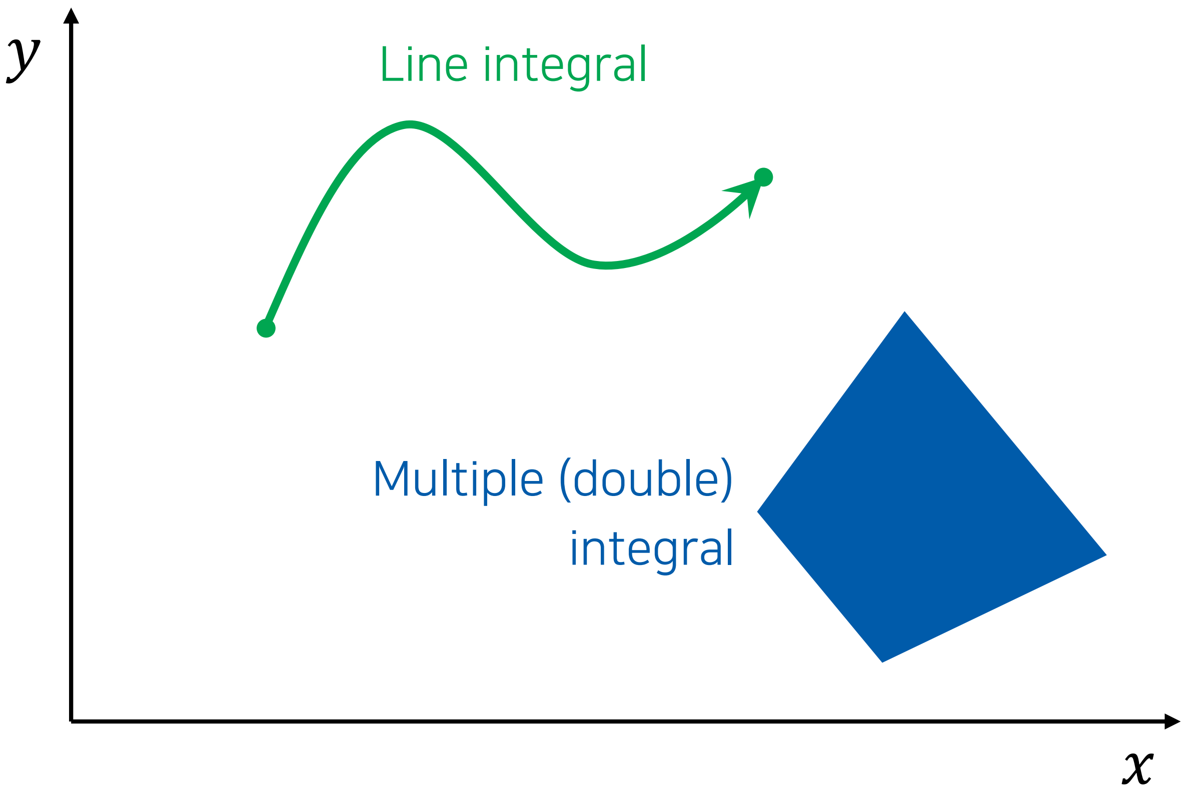
- Equilibrium state: a point in a space
- Reversible process: a line integral



오늘의 수업

- Multiple integrals
  - Double and triple integrals
  - In different coordinate systems
- Series
  - Mathematical series
  - Functional series
  - Power series

Line Integral vs. Multiple Integral



Double Integral (이중적분)

$$I = \underbrace{\int_{a_1}^{a_2} \underbrace{\int_{b_1}^{b_2} f(x, y)\, \mathrm{d}y}\, \mathrm{d}x}$$

- Carry out the inner integration first
  - Treat the other independent variable (x) as a constant
- Then carry out the outer integration

Example 9.6

Evaluate the double integral

$$I = \int_0^a \int_0^b (x^2 + 4xy)\,\mathrm{d}y\,\mathrm{d}x.$$

Carry out the inner integral first (x = constant):

$$\int_0^b (x^2 + 4xy)\,\mathrm{d}y = [x^2 y + 2xy^2]_0^b = x^2 b + 2xb^2$$

Now carry out the outer integral:

$$I = \int_0^a (x^2 b + 2xb^2)\,\mathrm{d}x = \left[\frac{b}{3}x^3 + b^2 x^2\right]_0^a = \frac{a^3 b}{3} + a^2 b^2$$

Example 9.7

Evaluate the double integral

$$\int_0^a \int_0^{3x} (x^2 + 2xy + y^2)\,\mathrm{d}y\,\mathrm{d}x.$$

Carry out the inner integral first (x = constant):

$$\int_0^{3x} (x^2 + 2xy + y^2)\,\mathrm{d}y = \left[x^2y + xy^2 + \frac{y^3}{3}\right]_0^{3x} = 3x^3 + 9x^3 + 9x^3 = 21x^3$$

Now carry out the outer integral:

$$\int_0^a 21x^3\,\mathrm{d}x = \left[\frac{21}{4}x^4\right]_0^a = \frac{21}{4}a^4$$

Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z)\, \mathrm{dz}\, \mathrm{d}y\, \mathrm{d}x$$

- First, integrate z from c_1 to c_2 (x, y = constant)
- Then integrate y from b_1 to b_2 (x = constant)
- Lastly, integrate x from a_1 to a_2

Example 9.9

For integers n, m, and k, find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

Carry out the innermost integral first (x, y = constant):

$$\int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z = \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \int_0^c \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z$$

From the integral table,

$$\int_0^c \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z = \frac{c}{2}$$

Example 9.9

For integers n, m, and k, find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

Next, treating x as a constant,

$$\int_0^b \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \frac{c}{2} \mathrm{d}y = \sin^2\left(\frac{n\pi x}{a}\right) \frac{c}{2} \int_0^b \sin^2\left(\frac{m\pi y}{b}\right) \mathrm{d}y$$

Similarly, from the integral table,

$$\int_0^b \sin^2\left(\frac{m\pi y}{b}\right) \mathrm{d}y = \frac{b}{2}$$

Example 9.9

For integers n, m, and k, find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

Lastly,

$$I = A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \frac{bc}{4} \mathrm{d}x = \frac{A^2 bc}{4} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \mathrm{d}x$$

Similarly, from the integral table,

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \mathrm{d}x = \frac{a}{2}$$

and we obtain $I = A^2 abc/8$.

Not in Mortimer

Separation of Variables

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x$$

If $f(x, y, z) = f_x(x) f_y(y) f_z(z)$,

$$I = \left[\int_{a_1}^{a_2} f_x(x)\, \mathrm{d}x\right] \times \left[\int_{b_1}^{b_2} f_y(y)\, \mathrm{d}y\right] \times \left[\int_{c_1}^{c_2} f_z(z)\, \mathrm{d}z\right]$$

Example 9.9 Revisited

For integers n, m, and k, find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

The integral can be decomposed as

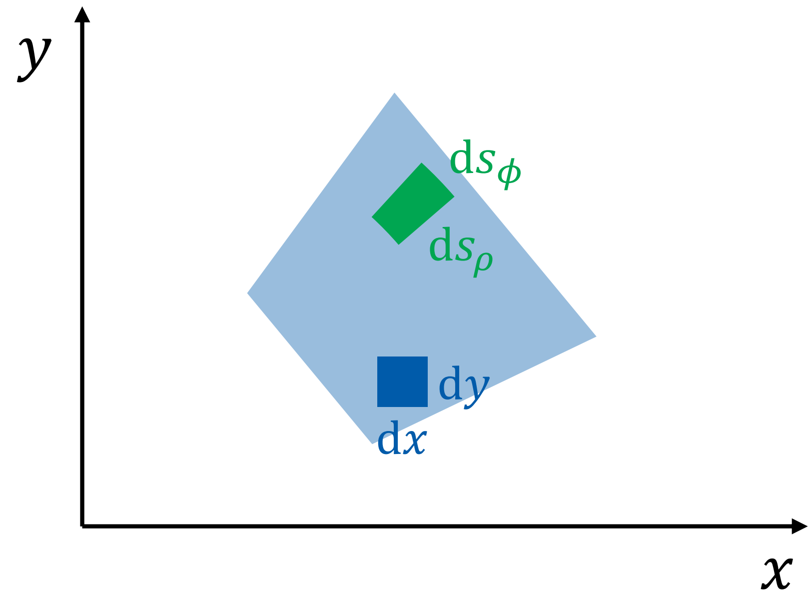
$$I = A^2 \left[\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \mathrm{d}x\right]\left[\int_0^b \sin^2\left(\frac{m\pi y}{b}\right) \mathrm{d}y\right]\left[\int_0^c \sin^2\left(\frac{k\pi z}{c}\right) \mathrm{d}z\right]$$

Hence, using the integral table,

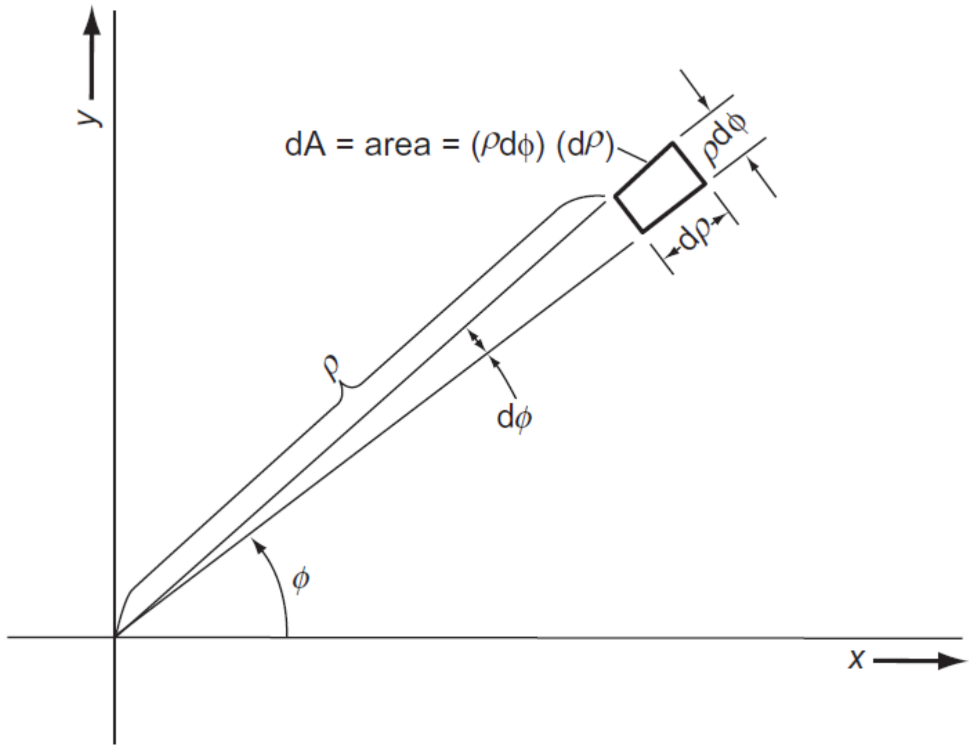
$$I = A^2 \times \frac{a}{2} \times \frac{b}{2} \times \frac{c}{2} = \frac{A^2 abc}{8}$$

Changing Variables in Double Integrals

$$\int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y \underset{?}{\rightarrow} \int\int f(\rho,\phi)\,\mathrm{d}\rho\,\mathrm{d}\phi$$



2-Dimensional Polar Coordinates



$$\mathrm{d}A = \mathrm{d}x\mathrm{d}y = \rho\mathrm{d}\rho\mathrm{d}\phi$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.4)

Conversion Formula

$$\int\int f\,\mathrm{d}A = \int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int\int f(\rho,\phi)\rho\,\mathrm{d}\rho\,\mathrm{d}\phi$$

Example 9.10

$$\psi = B\exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x-y plane is equal to unity.

The transformation, $x = \rho\cos\phi$ and $y = \rho\sin\phi$, leads to

$$\psi = B\exp[-a\{(\rho\cos\phi)^2 + (\rho\sin\phi)^2\}]$$

$$= B\exp[-a\rho^2(\cos^2\phi + \sin^2\phi)] = Be^{-a\rho^2}$$

Its square is

$$\psi^2 = B^2 e^{-2a\rho^2}$$

Example 9.10

$$\psi = B\exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x-y plane is equal to unity.

$$\psi^2 = B^2 e^{-2a\rho^2}$$

We require that

$$I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \psi^2 \,\mathrm{d}x\,\mathrm{d}y = 1$$

Substitution leads to

$$I = B^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-2a\rho^2}\,\mathrm{d}x\,\mathrm{d}y$$

Example 9.10

$$\psi = B\exp[-a(x^2+y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x-y plane is equal to unity.

$$I = B^2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-2a\rho^2}\,\mathrm{d}x\,\mathrm{d}y$$

$$= B^2\int_0^{2\pi}\int_0^{\infty} e^{-2a\rho^2}\rho\,\mathrm{d}\rho\,\mathrm{d}\phi$$

$$= B^2\int_0^{\infty} e^{-2a\rho^2}\rho\,\mathrm{d}\rho\int_0^{2\pi}\mathrm{d}\phi$$

$$I = 2\pi B^2\int_0^{\infty} e^{-2a\rho^2}\rho\,\mathrm{d}\rho$$

Example 9.10

$$\psi = B\exp[-a(x^2+y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x-y plane is equal to unity.

$$I = 2\pi B^2 \int_0^\infty e^{-2a\rho^2}\rho\,\mathrm{d}\rho$$

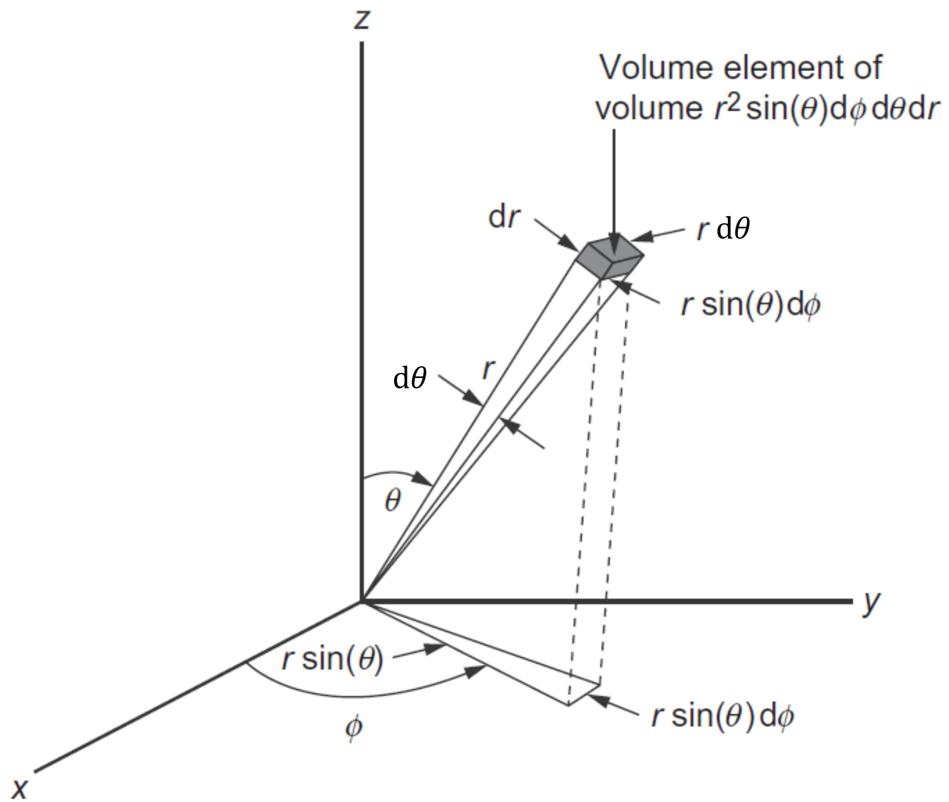
Use the method of substitution, letting $u = 2a\rho^2$ and $\mathrm{d}u = 4a\rho\mathrm{d}\rho$;

$$I = 2\pi B^2 \times \frac{1}{4a}\int_0^\infty e^{-u}\,\mathrm{d}u = \frac{\pi B^2}{2a}$$

Since $I = 1$, $B^2 = 2a/\pi$ and

$$B = \sqrt{2a/\pi}$$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin\theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Example 9.12

Evaluate the integral

$$I = \int_0^3 \int_0^\pi \int_0^{2\pi} r \cos^2 \phi \, r^2 \sin \theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}r.$$

The integral can be decomposed:

$$I = \int_0^3 r^3 \, \mathrm{d}r \int_0^\pi \sin \theta \, \mathrm{d}\theta \int_0^{2\pi} \cos^2 \phi \, \mathrm{d}\phi$$

$$\int_0^3 r^3 \, \mathrm{d}r = \left[\frac{1}{4} r^4\right]_0^3 = \frac{81}{4}, \qquad \int_0^\pi \sin \theta \, \mathrm{d}\theta = [-\cos \theta]_0^\pi = 2$$

$$\int_0^{2\pi} \cos^2 \phi \, \mathrm{d}\phi = \int_0^{2\pi} \frac{\cos(2\phi) + 1}{2} \mathrm{d}\phi = \left[\frac{1}{4} \sin(2\phi) + \frac{1}{2}\phi\right]_0^{2\pi} = \pi$$

Hence, $I = 81\pi/2$.

10. Mathematical Series

(10장 급수)

Sequence and Series

- Sequence (수열): $a_0 \to a_1 \to a_2 \to \cdots \to a_n \to \cdots$
  - Rule for generating one member from previous one

- Series (급수): $a_0 + a_1 + a_2 + \cdots$
  - Finite series (유한급수): $s = a_0 + a_1 + \cdots + a_n$
  - Infinite series (무한급수): $s = a_0 + a_1 + \cdots + a_n + \cdots$

More about Infinite Series

$$s = a_0 + a_1 + \cdots + a_n + \cdots = \sum_{k=0}^{\infty} a_k$$

- nth partial sum (부분합): $S_n = a_0 + a_1 + \cdots + a_{n-1}$
  - $s = \lim_{n \to \infty} S_n$

- If s exists and is finite, the series converges; otherwise, the series diverges.

Geometric Series (기하급수)

$$s = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = a \sum_{n=0}^{\infty} r^n$$

- Convergence condition: $|r| < 1$

- Value: since $s = a + r(a + ar + ar^2 + \cdots) = a + rs$,

$$s = \frac{a}{1-r}$$

Example 10.3

The molecular partition function z is defined as

$$z = \sum_{i=0}^{\infty} \exp\left(-\frac{E_i}{k_{\mathrm{B}}T}\right).$$

If we consider only the vibration of a diatomic molecule, to a good approximation,

$$E_i = E_v = h\nu\left(v + \frac{1}{2}\right).$$

Find an expression of the vibrational partition function.

Example 10.3

$$z = \sum_{v=0}^{\infty} \exp\left(-\frac{E_v}{k_{\mathrm{B}}T}\right), \qquad E_v = h\nu\left(v + \frac{1}{2}\right)$$

Substitution leads to

$$z = \sum_{v=0}^{\infty} \exp\left[-\frac{h\nu}{k_{\mathrm{B}}T}\left(v + \frac{1}{2}\right)\right] = e^{-h\nu/2k_{\mathrm{B}}T} \sum_{v=0}^{\infty} e^{-h\nu v/k_{\mathrm{B}}T}$$

Introduce $r = e^{-h\nu/k_{\mathrm{B}}T}$ to yield

$$z = r^{1/2} \sum_{v=0}^{\infty} r^{v} = r^{1/2} \frac{1}{1-r} = \frac{e^{-h\nu/2k_{\mathrm{B}}T}}{1 - e^{-h\nu/k_{\mathrm{B}}T}}$$

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

- Coefficients (계수): $a_0, a_1, a_2, \cdots$
- Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$

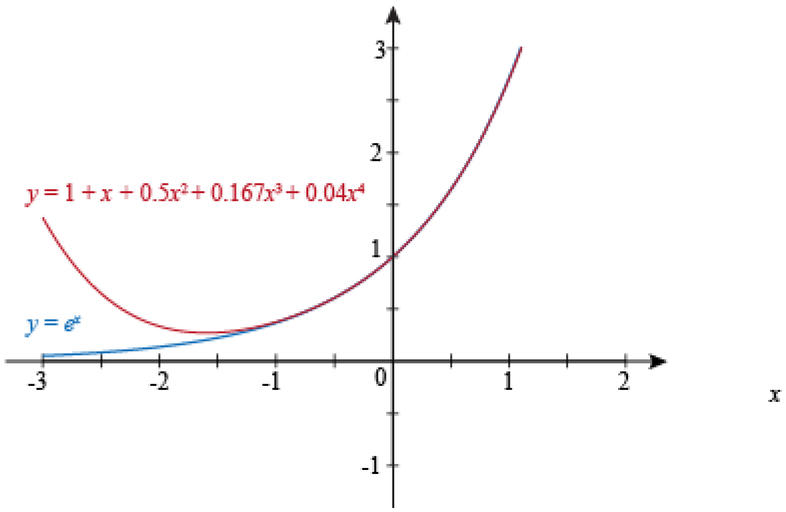
1. Power series (멱급수): powers as basis
2. Fourier series (푸리에 급수): sines & cosines as basis

Approximation to a Function

- Functional series can be used to represent a function

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots \to f(x)$$

- Its partial sum can be used as an approximation



(Image: https://www.shmoop.com/series/taylor-maclaurin-series-examples.html)

Power Series

- Maclaurin Series

$$s(x) = a_0 + a_1 x + a_2 x^2 + \cdots$$

- Taylor Series

$$s(x) = a_0 + a_1(x-h) + a_2(x-h)^2 + \cdots$$

Maclaurin Series to Represent $\boldsymbol{f(x)}$

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$$

At $x = 0$, $f(0) = a_0$ and

$$\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=0} = n!\, a_n$$

Hence,

$$a_n = \frac{1}{n!}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=0}$$

Example 10.7

Find all of the coefficients for the Maclaurin series representing $\sin x$.

Consider $x = 0$. Since $\sin 0 = 0$, $a_0 = 0$.

For $f(x) = \sin x$,

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \cos x\,, \qquad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = -\sin x\,, \qquad \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = -\cos x\,, \qquad \frac{\mathrm{d}^4 f}{\mathrm{d}x^4} = \sin x\,, \cdots$$

Hence, using $a_n = \frac{1}{n!}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=0}$,

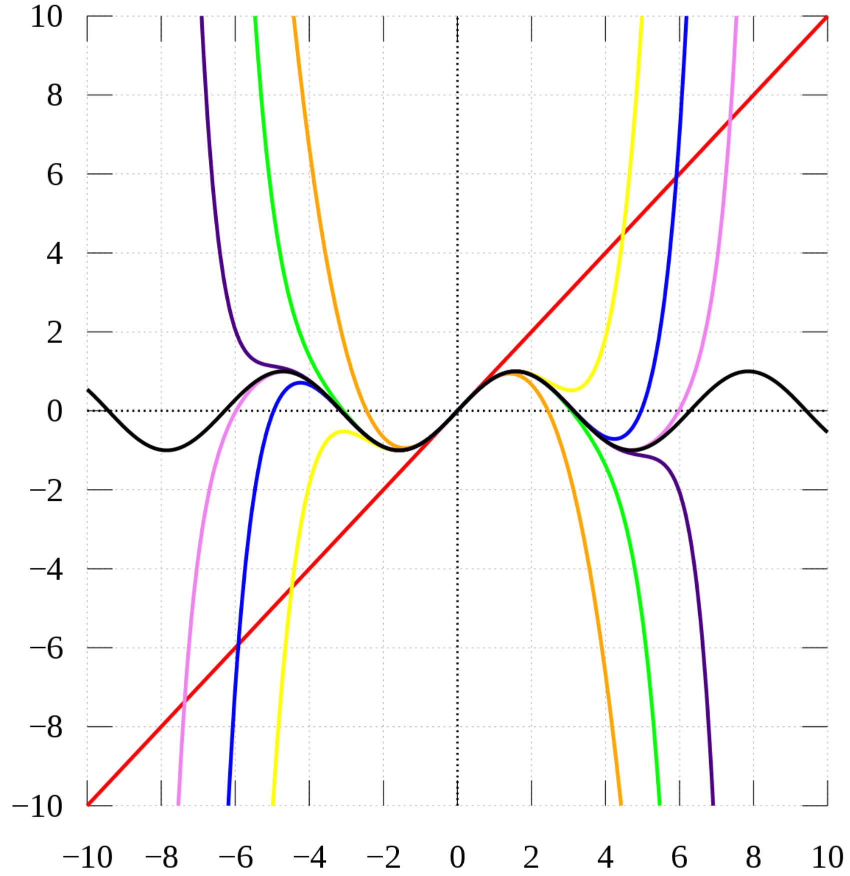
$$a_1 = 1,\ a_2 = 0,\ a_3 = -1/3!\,,\ a_4 = 0, \cdots$$

and

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots$$

Not in Mortimer

Maclaurin Approximation to Sine



(Image: https://commons.wikimedia.org/wiki/File:Sintay_SVG.svg)

Taylor Series to Represent $\boldsymbol{f(x)}$

$$f(x) = a_0 + a_1(x-h) + a_2(x-h)^2 + \cdots$$

where $h \neq 0$.

$$a_0 = f(h), \qquad a_n = \frac{1}{n!}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=h}$$

- Often called Taylor expansion about $x = h$
- A function that is not differentiable at $x = 0$ can be represented by a Taylor series

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

The series is given as

$$\ln x = a_0 + a_1(x-1) + a_2(x-1)^2 + a_3(x-1)^3 + a_4(x-1)^4 \cdots$$

where

$$a_0 = f(1), \qquad a_n = \frac{1}{n!}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=1}$$

For $f(x) = \ln x$, $a_0 = \ln 1 = 0$.

Also,

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{1}{x}, \qquad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = -\frac{1}{x^2}, \qquad \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = \frac{2}{x^3}, \qquad \frac{\mathrm{d}^4 f}{\mathrm{d}x^4} = -\frac{3!}{x^4}, \cdots$$

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{1}{x}, \qquad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = -\frac{1}{x^2}, \qquad \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = \frac{2}{x^3}, \qquad \frac{\mathrm{d}^4 f}{\mathrm{d}x^4} = -\frac{3!}{x^4}, \cdots$$

Hence,

$$a_1 = \frac{1}{1!}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)_{x=1} = 1, \qquad a_2 = \frac{1}{2!}\left(\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right)_{x=1} = -\frac{1}{2},$$

$$a_3 = \frac{1}{3!}\left(\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right)_{x=1} = \frac{1}{3}, \qquad a_4 = \frac{1}{4!}\left(\frac{\mathrm{d}^4 f}{\mathrm{d}x^4}\right)_{x=1} = -\frac{1}{4}, \cdots$$

Therefore,

$$\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \cdots$$

Change of Variable

$$f(x) = a_0 + a_1(x-h) + a_2(x-h)^2 + \cdots$$

can be written as

$$f(x+h) = a_0 + a_1 x + a_2 x^2 + \cdots$$

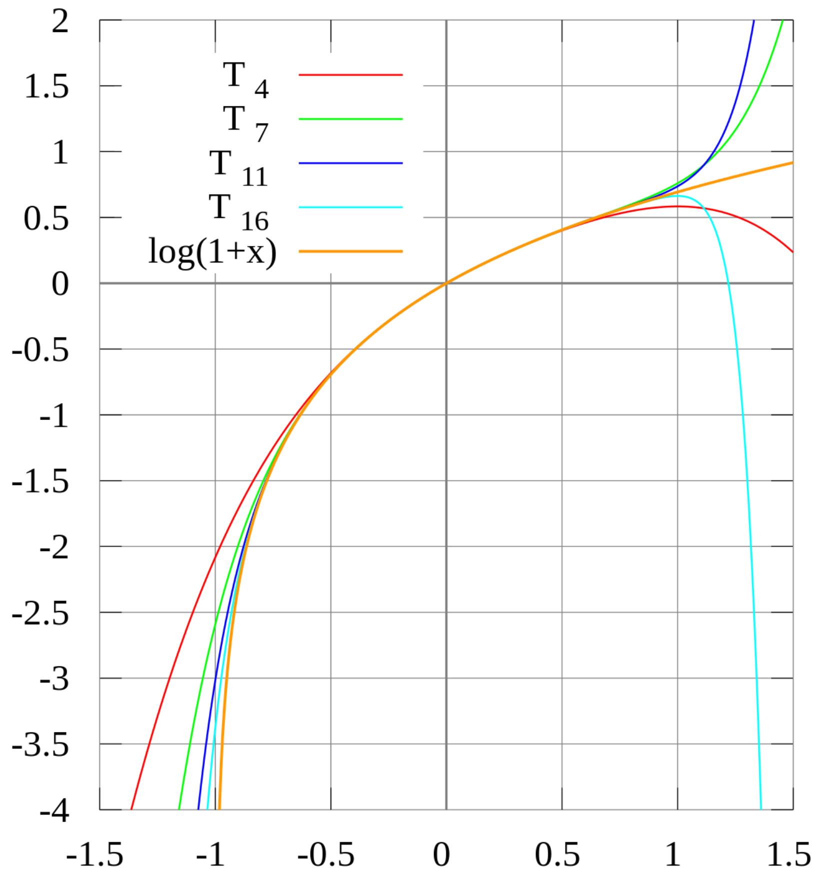
$$a_0 = f(x=0), \qquad a_n = \frac{1}{n!}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)_{x=0}$$

e.g. Taylor expansion of $\ln(x+1)$:

$$\ln(x+1) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots$$

Not in Mortimer

Taylor Approximation to $\ln(x + 1)$



(Image: https://commons.wikimedia.org/wiki/File:LogTay.svg)

Mathematical Operations on Series

Differentiation/integration of a series?

For a (uniformly convergent) functional series

$f(x) = \sum_{n=0}^{\infty} a_n g_n(x)$,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left[\sum_{n=0}^{\infty} a_n g_n(x)\right] = \sum_{n=0}^{\infty} a_n \frac{\mathrm{d}g_n(x)}{\mathrm{d}x}$$

$$\int_b^c f(x)\,\mathrm{d}x = \int_b^c \left[\sum_{n=0}^{\infty} a_n g_n(x)\right] \mathrm{d}x = \sum_{n=0}^{\infty} a_n \int_b^c g_n(x)\,\mathrm{d}x$$

Example 10.12

Find the Maclaurin series for $\cos x$ from the Maclaurin series for $\sin x$.

The Maclaurin series for $\sin x$ is

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots.$$

Since $\mathrm{d}(\sin x)/\mathrm{d}x = \cos x$, differentiating both sides,

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x,y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + a_{21}x^2y + a_{12}xy^2 + \cdots$$

$x = 0, y = 0$ gives $f(0,0) = a_{00}$

What about other coefficients?

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + a_{21}x^2y + a_{12}xy^2 + \cdots$$

$$(\partial f/\partial x)_y = a_{10} + a_{11}y + 2a_{20}x + 2a_{21}xy + a_{12}y^2 + \cdots$$

$x = 0, y = 0$ gives $(\partial f/\partial x)_y\big|_{0,0} = a_{10}$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2$$
$$+ a_{21}x^2y + a_{12}xy^2 + \cdots$$

$$(\partial f/\partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \cdots$$

$x = 0, y = 0$ gives $(\partial f/\partial y)_x|_{0,0} = a_{01}$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2$$
$$+ a_{21}x^2y + a_{12}xy^2 + \cdots$$

$$(\partial f/\partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \cdots$$
$$(\partial^2 f/\partial x \partial y) = a_{11} + 2a_{21}x + 2a_{12}y + \cdots$$

$x = 0, y = 0$ gives $(\partial^2 f/\partial x \partial y)|_{0,0} = a_{11}$

Power Series in Several Variables

For a two-variable function $f(x,y)$, we can write

$$f(x,y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + a_{21}x^2y + a_{12}xy^2 + \cdots$$

In general,

$$a_{nm} = \frac{1}{n!\,m!}\left(\frac{\partial^{n+m}f}{\partial^m y\,\partial^n x}\right)\Bigg|_{0,0}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x,y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y.

Consider $f(x,y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + \cdots$

$$f(0,0) = e^0 = 1 = a_{00}$$

$$(\partial f/\partial x)_y\Big|_{0,0} = e^{x+y}\Big|_{0,0} = e^0 = 1 = a_{10}$$

$$(\partial f/\partial y)_x\Big|_{0,0} = e^{x+y}\Big|_{0,0} = e^0 = 1 = a_{01}$$

All the derivatives are equal to e^{x+y} and

$$a_{nm} = \frac{1}{n!\,m!}\left(\frac{\partial^{n+m} f}{\partial^m y \partial^n x}\right)\Bigg|_{0,0} = \frac{1}{n!\,m!}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x, y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y.

$$e^{x+y} = 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \cdots$$

The Maclaurin series representing e^x is

$$e^x = 1 + x + (1/2!)x^2 + (1/3!)x^3 + \cdots$$

Similarly,

$$e^y = 1 + y + (1/2!)y^2 + (1/3!)y^3 + \cdots$$

Hence, the product of e^x and e^y is

$$e^{x+y} = (1 + x + (1/2!)x^2 + \cdots)(1 + y + (1/2!)y^2 + \cdots)$$

$$= 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \cdots$$